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Windshield phone mount with extension arm bumper

Abstract

A windshield phone mount includes a telescoping arm assembly that brings the mounted cell phone or other personal electronic device within the maximum reach envelope of the driver. A bumper substantially surrounds an arm component in cross section and rests on the dashboard near the rear edge thereof. Living hinges in the bumper allow it to be pried apart and installed on the arm component without tearing. A user may select from a kit of arms for use in assembling the complete mount.

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Background/Summary

RELATED APPLICATIONS (1) This application is a continuation in part of copending U.S. patent application Ser. No. 17/895,779, owned by the applicant hereof. The entire disclosure and drawings of the foregoing application are hereby fully incorporated by reference herein.

BACKGROUND OF THE INVENTION

(1) It is known to mount a personal electronic device, such as a cell phone, a tablet or a GPS device, in a vehicle interior. Mounts have been provided to mount the device to a variety of interior surfaces, such as a cupholder, a heater/air conditioning vent, a dashboard or the inside surface of the vehicle windshield.

(2) Recent models of these devices have displays that take up the entire front of the device. Ideally, the front surface should be presented to the driver in a plane orthogonal to the driver's line of sight. Further, the lens of the human eye changes shape when it changes focus from an object closer to the eye to an object farther away from it. When driving, what is happening out on the road is close to infinity in terms of lens focus. To minimize the disparity in focus, the device could be positioned on the windshield to be as far from the driver as possible. It is also advantageous to position any such display such that the driver does not have to divert his or her line of sight from what is happening on the road. But, for many such devices, it is advantageous to position the device within arm's reach of the driver.

(3) This in turn means that the device should be mounted close to the interior surface of the windshield, not far from the driver's forward line of sight. In most current vehicles, the windshield slants upwardly and rearwardly by a considerable angle. Mounting the device close to the windshield interior surface effectively obstructs the removal of the device from the mount in a vertical direction, and likewise prohibits the reinstallation of the device into the mount from this direction.

(4) Ideally, a device should be able to be installed into the mount in a horizontal direction, and removed from the mount the same way. Further, it would be ideal to install the device into, and remove the device from, the mount with one hand.

(5) One consideration in mounting a cell phone in a vehicle is whether the device as mounted is within a hand reach control envelope relative to the driver. Positioning a cell phone or other personal electronic device rearwardly of a dashboard margin places the cell phone within this hand reach control envelope. A need exists to provide such a mount, in such a way that the cell phone and the mount stay affixed to vehicle support surfaces and are resistant to shear forces tending to separate them.

SUMMARY OF THE INVENTION

(6) According to one aspect of the invention, a mount for a personal electronic device is provided. The mount, which can be for a cell phone, a tablet, a GPS device or the like, is configurable to assume an open configuration in which the device may be removed from the mount in a predetermined first direction, such as rearward, and a closed configuration in which the device may not be so removed in that direction. The mount has first and second support plates operable to support the device. The first support plate is substantially oriented in a first plane when the mount is in the closed configuration, and is substantially oriented in a second plane at an angle to the first plane in when the mount is in the open configuration. In either configuration, the second support plate stays substantially oriented in the first plane.

(7) In one embodiment, the first support plate is part of a first jaw and the second support plate is part of a second jaw. The first jaw has a laterally inwardly facing first channel. The second jaw has

a laterally inwardly facing second channel. The first channel is adapted to cage a first lateral side of the device when the mount is in the closed configuration. The second channel is adapted to cage a second lateral side of the device, with the second lateral side being laterally spaced from the first lateral side. The second jaw is laterally slidable relative to a base of the mount so as to size a width between the first channel and the second channel of the mount to a width of the device between its first and second lateral sides.

(8) In one embodiment, the first support plate is a portion of a first jaw. The first jaw has a first channel surface which rearwardly extends from the first support plate to a rearward end. A first lip of the first jaw laterally inwardly extends from the rearward end of the first channel surface. The first support plate and the first lip are operable to cage the first lateral side of the device when the mount is in the closed configuration. In the open configuration, the first lip is laterally outwardly displaced, such that the first lateral side of the device is no longer caged between the first support plate and the first lip. This permits the device to be removed from the mount in the first direction.

(9) In another aspect of the invention, a mount for a personal electronic device is provided, wherein the mount is configurable to an open configuration in which the device may be removed from the mount in a predetermined first direction, such as towards the rear, and is further configurable to assume a closed configuration in which the device may not be removed from the mount in the first direction. The mount has a base with a laterally outwardly extending first finger tab, and a holding element hinged to the base and having a laterally outwardly extending second finger tab. The second finger tab is spaced from the first finger tab by a first distance when the mount is in the open configuration. The second finger tab is positioned to be farther away than the first distance from the first finger tab when the mount is in the closed configuration. A user may pinch the first and second finger tabs, moving the second finger tab toward the first finger tab, in order to reconfigure the mount from the closed configuration to the open configuration.

(10) In another aspect of the invention, a mount for a personal electronic device is configurable to assume an open configuration in which the device may be removed from the mount in a predetermined first direction, such as rearward, or to assume a closed configuration in which the device may not be removed from the mount in the first direction. The mount has a base and a first support plate hinged to the base at an axis. At least one tab of the first support plate extends radially inwardly relative to the axis. The first support plate substantially occupies a first plane when the mount is in the closed configuration, and substantially occupies a second plane when the mount is in the open configuration. The second plane is at an angle to the first plane. The tab is operable to contact a back of the device when it is desired to install the device into the mount in a second direction opposite the first direction. A predetermined force exerted by the user through the back of the device will cause the first support plate to pivot from the second plane to the first plane, thereby causing the mount to be reconfigured from the open configuration to the closed configuration.

(11) In one embodiment, the mount has a second support plate that has an inner margin. The second support plate is substantially aligned with the first plane. The second support plate has a laterally extending slot. A fastener extends in the first direction from the base and into the slot, such that the second support plate is slidably affixed to the base. The second support plate is laterally slidable from a narrowest position in which the second support plate is closest to the first support plate, to a widest position in which the second support plate is farthest away from the first support plate. In the narrowest position, the tab of the first support plate abuts an inner margin of the second support plate.

(12) In one embodiment, the aforementioned tab is one of an upper and lower tab on the first support plate. A center portion of the right margin of the second support plate extends more to the right than does either an upper portion or a lower portion of this margin, so as to extend the slot, and therefore the lateral distance through which the second support plate can slide. When mated together at a minimum width position, the center portion of the second support plate right margin is disposed rightward of the upper and lower inner tabs of the first support plate.

(13) In one embodiment, the first support plate is operable to be moved from the first plane to the second plane, and vice versa. A detenting structure indexes the first and second planes, so that the mount snaps open and closed. The detenting structure resists up to a predetermined degree of rotational force, to keep the mount in the desired open or closed configuration, and provides tactile and aural feedback to the user confirming that the desired configuration has been achieved.

(14) In another aspect of the invention, a mount for a personal electronic device includes a base that is substantially oriented to a base plane. A pivoting jaw is hinged to the base at an axis. The axis is disposed parallel to the base plane. The pivoting jaw is pivotable between a closed configuration in which the device may not be removed from the mount in a predetermined first direction, such as rearward, and an open configuration in which the device may be so removed. The pivoting jaw has a side plate extending in the first direction to a side plate end. A lip of the pivoting jaw extends laterally inwardly from the side plate end by a lip width to an inner lip end. The inner lip end is at a predetermined radius from the axis, where the radius is orthogonal to the axis. A position of the inner lip end while in the open configuration is separated from a position of the inner lip end while in the closed configuration by an angle measured around the axis. A ratio of the lip width to the radius is no greater than a sine of the angle, such that in the open configuration, the inner lip end will be angularly displaced out of the way from a path of a lateral side of the electronic device as the device is being removed from the mount in the first direction.

(15) In another aspect of the invention, a method for mounting a personal electronic device to a surface is provided. The surface may, for example, be an interior surface of a slanting vehicle windshield. A mount is provided that has a closed configuration in which the device may not be removed from the mount in a predetermined first direction (such as rearward), and an open configuration in which such removal is possible. The mount further has a base and an attachment arm with a first end fastened to the base and a second end remote from the base.

(16) The mount is either received by the user in the closed configuration, or the user places the mount in the closed configuration. Next, the device is placed on the first and second support plates of the mount. The second support plate is slid in a first plane relative to the base, so that a channel rearwardly extending from the second support plate cages a second lateral side of the device, and so that a channel rearwardly extending from the first support plate cages a first lateral side of the device, with the second lateral side being spaced from the first lateral side. Next, the device is removed from the mount, in a second direction substantially orthogonal to the first direction, such as upward. Alternatively the user may pinch the external tabs of the first jaw and the base to put the first support plate in an open configuration, thereby permitting the electronic device to be removed from the mount in a rearward direction.

(17) The now-empty but width-adjusted mount is next attached, by the second end of the attachment arm, to the surface. If he or she has not already done so, the user then pivots the first support plate relative to the base, such that the first support plate is substantially aligned to a second plane that is tilted relative to the first plane. This configures the mount to the open configuration. The user then moves the device in a third direction which is opposite the first direction (e.g. forward) until the back of the device contacts the first support plate. The user further moves the device in the third direction, causing the first support plate to pivot from the second plane back to the first plane, thereby caging the first lateral side of the device and preventing the device's removal from the mount in the first direction. This securely holds the electronic device relative to the windshield.

(18) Later, the user may wish to remove the device from the mount. This is accomplished by pinching a first finger tab, which laterally outwardly extends relative to the first support plate, toward a second finger tab, which laterally outwardly extends from the base. This pivots the first support plate to the second plane, and causes the first lateral side of the device to be uncaged from the first channel, permitting the device to be removed from the mount in the first direction.

(19) The mount may therefore hold the electronic device in close proximity to the inner surface of

the windshield, and in a position in which it would not be possible to lift the device upwardly out of the jaws of the mount. With one hand, the user pinches together the finger tabs, opening the mount and permitting the removal of the device from the mount. To install the device, the user simply presses, with one hand, the back of the device against the inner tab(s) of the pivoting support plate, causing the first support plate to pivot back to the closed configuration. As so installed, the device is closer to the focal plane that the driver is using while driving on the road, enhancing the acuity with which the driver perceives road conditions and decreasing any loss of focus on the road caused by focusing on the device screen.

(20) In another aspect of the invention, a mount for a personal electronic device, such as a cell phone, GPS device, tablet or the like, has a mounting component for affixation to a rear surface of a vehicle windshield. The mount further has a device mounting bracket for holding the personal electronic device. An elongate arm assembly is joined to and extends between the mounting component and the device mounting bracket. The arm assembly has at least a first arm component that, in use, is positioned over the rearward margin of a vehicle dashboard. At least a portion of this first arm component has a predetermined cross-sectional shape. A bumper, formed of elastomeric material, substantially surrounds the cross-sectional shape of the first arm component. The bumper is positioned on the first arm component so as to be near the rearward margin of the vehicle dashboard, such that the bumper, and not the first arm component, will contact the dashboard.

(21) In one embodiment, the bumper is integrally molded of an elastomeric material. The bumper has a hollow inner shape which substantially conforms to the cross-sectional shape of the first arm component, a bumper outer periphery spaced from and surrounding the inner shape, and an opening extending from the bumper outer periphery to the inner shape. The opening has first and second opposed sidewalls that are disposed at a first width from each other when the bumper is in an unflexed condition. At least one outer groove inwardly extends toward the inner shape from the bumper periphery, and has a groove bottom near the inner shape. At least one living hinge is formed between the groove bottom and the inner shape. Prying force flexes this living hinge and moves the first sidewall away from the second sidewall, such that the first sidewall is positioned at second width from the second sidewall, where the second width is greater than the first width. This permits the insertion of the cross-sectional shape of the first arm component through the opening and into the inner shape of the bumper. When prying force is removed, the first sidewall is again positioned to be at the first width from the second sidewall. In some embodiments, several living hinges of the bumper are created by the near approach of a plurality of grooves to the inner shape, aiding in the flexing of the bumper.

(22) In embodiments, the aforementioned arm assembly also includes a second arm component. The first arm component is slidable with respect to the first arm component such that a length of the arm assembly between the mounting component and the device mounting bracket may be adjusted. In some embodiments, the arm assembly further includes a third arm component that is slidably attached to the second arm component. A forward end of the third arm component may be joined to the mounting component, while a rearward end of the first arm component may be joined to the device mounting bracket.

(23) In a further aspect of the invention, there is provided a kit for mounting a personal electronic device to a rear surface of a vehicle windshield. The kit includes a mounting component for affixation to the rear surface of a vehicle windshield, a device mounting bracket for holding the personal electronic device, and a plurality of elongate arms each adapted to join the mounting component to the device mounting bracket. The arms include a first arm and a second arm. The first arm has a first rear end adapted to be rotatably joined to the mounting component, and a first forward end remote from the first rear end and adapted to be rotatably joined to the device mounting bracket. The second arm has a second rear end adapted to be rotatably joined to the mounting component and a second forward end remote from the second rear end. The second forward end is separated from the second rear end by a second total length that is greater than the

first length. The user selects either the first or the second arm for assembly to the mounting component and to the device mounting bracket, thereby completing a mount for mounting the electronic device to the rear surface of the windshield.

(24) In one embodiment, the second arm has a plurality of second arm components slidable in relation to each other and affixed to each other in a user-selected position, such that the second total length may be adjusted by the user. In one embodiment, the kit may further include a third arm whose length is intermediate the first arm length and the second arm total length.

(25) In a further aspect of the invention, a bumper is provided for an arm assembly of a cell phone mount for mounting to a rear surface of a windshield of a vehicle, wherein the arm assembly further is adapted to place the cell phone rearwardly of a vehicle dashboard. A body of the bumper is formed of an elastomer. The body has a hollow inner shape and an outer periphery spaced from the inner shape. The inner shape substantially surrounds a cross section of the arm assembly. The outer periphery has an upper portion and a lower portion spaced from the upper portion. The upper and lower portions of the bumper each have a plurality of outwardly extending fingers separated from each other by grooves, such that, in use, either the fingers of the upper portion or the fingers of the lower portion will cushion contact of the arm assembly with the vehicle dashboard.

(26) In a still further aspect of the invention, a bumper and arm assembly is provided for a cell phone mount. The assembly includes an elongate arm having a length between a forward end and a rearward end. The arm has a uniform cross section along a major portion of its length. The bumper is formed of an elastomeric material and has a hollow, cross-sectional inner shape that substantially conforms to the uniform cross-section of the major portion of the arm. The bumper further has a cross-sectional outer shape. An opening of the bumper joins the inner shape to the outer shape and separates first and second ends of the bumper. At least one hinge is formed in the bumper between the inner shape and the outer shape, such that a user may pry apart the first and second ends of the bumper, flexing the hinge. The user then fits the bumper around the arm at any point along the major portion of the length of the arm, such that the inner shape of the bumper substantially conforms to the uniform cross-section of the arm.

(27) In another aspect of the invention, a bumper is provided for supporting an elongate structural member above a support surface. The bumper has a body integrally molded of an elastomeric material. A hollow inner shape is formed in the body. This inner shape substantially conforms to a predetermined cross-sectional shape of the structural member. An outer periphery of the body substantially surrounds the inner shape in cross section and is spaced from it. The outer periphery contacts the support surface when the bumper is mounted around the structural member and when the bumper is being used to support the structural member above the support surface. The bumper has a plurality of outer grooves which open on the outer periphery and extend inwardly toward the inner shape. At least one living hinge is formed between one of the bottoms of these grooves and the inner shape. An opening in the body extends from the outer periphery to the inner shape. The opening spaces a first side of the body from a second side of the body when the body is in an unflexed condition. The bumper is adapted to be installed around the cross-sectional shape of the structural member by exerting prying force between the first and second sides of the body, forcing the first side away from the second side. The living hinge flexes responsive to this prying force. The cross-sectional shape of the structural member is then inserted through the opening and is seated within the inner shape of the body, the first side of the body returning to a position at the first width from the second side when the prying force is removed.

(28) In an embodiment, there are multiple living hinges disposed in series between the first side of the body and the second side. For example, the inner shape may have a bottom opposed to the opening, a left side wall upwardly extending from the bottom and a right side wall upwardly extending from the bottom so as to be spaced from the left side wall. A left inner groove of the inner shape leftwardly extends from the left side wall toward the outer periphery and has a bottom. A right inner groove of the inner shape rightwardly extends from the right side wall toward the

outer periphery and has a bottom. The outer grooves include a first outer groove having a bottom near the bottom of the left inner groove, and a second outer groove having a bottom near the bottom of the right inner groove. A first living hinge is formed by the proximity of the bottoms of the left inner groove and the first outer groove. A second living hinge is formed by the proximity of the bottoms of the right inner groove and the second outer groove. In one embodiment, the body further has formed therein third and fourth living hinges. The third living hinge is formed by the proximity of a bottom of a third outer groove to the left inner groove bottom, and the fourth living hinge is formed by the proximity of a bottom of a fourth outer groove to the right inner groove bottom.

(29) In one embodiment, the outer periphery of the bumper has an upper portion. The outer grooves of the body include a central outer groove that extends downwardly from the upper portion toward the bumper inner shape. The inner shape has a downwardly extending finger which separates a left upper surface from a right upper surface. A left upper corner of the inner shape is formed by the left upper surface and the finger. A right upper corner of the inner shape is formed by the right upper surface and the finger. A first living hinge is formed between the bottom of the central groove and the left upper corner, while a second living hinge is formed between the bottom of the central groove and the right upper corner.

(30) The one or more living hinges of the body permit the distortion of the elastomeric bumper body when the structural member is being inserted through the body opening and into the inner shape, without tearing.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Further aspects of the invention and their advantages can be discerned in the following detailed description as read in conjunction with the drawings of an exemplary embodiment, in which like characters denote like parts and in which:

(2) FIG. 1 is a perspective view of a mount according to the invention, from a rear and side direction, jaws of the mount being in an intermediate position relative to each other, a first jaw being rotated to an open or release configuration;

(3) FIG. 2 is a top view of the mount as configured in FIG. 1;

(4) FIG. 3 is a perspective right side view of a mount according to the invention, as affixed to a slanting windshield and as holding a cell phone, the first jaw being rotated to a closed configuration;

(5) FIG. 4 is an exploded view of the mount shown in FIGS. 1-3;

(6) FIG. 4A is a rear view detail of a base plate as disassembled from the mount shown in FIGS. 1-3;

(7) FIG. 5 is a rear view of the mount shown in FIG. 1, with first and second jaws defining a minimum width between them, and the first jaw being rotated to a closed configuration;

(8) FIG. 6 is a top view of the mount as configured in FIG. 6;

(9) FIG. 7 is a rear view of the mount shown in FIG. 1, with first and second jaws in a maximum width position relative to each other, and the first jaw rotated to a closed configuration;

(10) FIG. 8 is a top view of the mount as configured in FIG. 7;

(11) FIG. 9 is a top view detail of the mount in which first and second jaws are in a maximum width position relative to each other, and wherein the first jaw has been rotated to an open or release configuration, a portion of the first jaw in the closed configuration being drawn in dotted line;

(12) FIG. 10 is a horizontal sectional view taken substantially along Line 10-10 of FIG. 7;

(13) FIG. 11 is a horizontal sectional view taken on the same plane as that used for FIG. 10, but

showing the first jaw as rotated to an open or release configuration;

(14) FIG. 12 is a front perspective detail of the first jaw prior to assembly to a base plate of the mount;

(15) FIG. 13 is a top view detail of the mount as holding a cell phone, the first jaw having been rotated to a closed configuration;

(16) FIG. 14 is a top view detail similar to FIG. 13, but showing the first jaw as rotated to an open or release configuration;

(17) FIG. 15A is a view of initial steps of installing a cell phone in the mount, the initial steps including sizing the mount for a particular cell phone;

(18) FIG. 15B is a view of a step of removing the cell phone from the mount after the jaws have been sized to the cell phone width, and also is a view of a step of removing the cell phone from the mount after the mount has been affixed to the windshield;

(19) FIG. 15C is a view of a step of removing a cell phone from the mount with one hand;

(20) FIG. 16 is a block diagram of steps of a process for installing a cell phone in a mount according to the invention, and affixing the mount to the interior surface of a windshield;

(21) FIG. 17 is a block diagram of steps of a process for removing a cell phone from the mount after the mount has been affixed to a windshield;

(22) FIG. 18 is a perspective detail, taken from above, of attachment arm and suction cup assemblies for use with the invention;

(23) FIG. 19 is a perspective detail, taken from below, of the attachment arm and suction cup assemblies shown in FIG. 18;

(24) FIG. 20 is a sectional view taken substantially along an axis of the attachment arm and suction cup assemblies shown in FIGS. 18 and 19;

(25) FIG. 21 is a schematic elevational view of a front portion of a vehicle interior, showing a maximum reach envelope of a driver and the use of a further embodiment of a windshield phone mount according to the invention;

(26) FIG. 22 is an elevational view of a windshield phone mount as including a telescoping extension arm and bumper;

(27) FIG. 23 is a top rear left perspective view of a windshield phone mount as installed in a vehicle;

(28) FIG. 24 is a bottom perspective view of forwardmost and intermediate arm components of the telescoping arm assembly;

(29) FIG. 25 is a bottom perspective view of intermediate and rearwardmost arm components of the telescoping arm assembly;

(30) FIG. 26 is an exploded top rear left perspective view of a telescoping arm assembly;

(31) FIG. 27 is a top left front perspective view of a rearmost arm segment of the telescoping arm assembly shown in FIGS. 23 and 24, further showing a bumper fitted to this arm segment;

(32) FIG. 28 is a sectional view taken substantially along Line 28-28 of FIG. 27;

(33) FIG. 29 is a sectional view similar to FIG. 28, but showing the bumper being pried apart by the rearwardmost arm segment as the bumper is being installed around the rearwardmost arm segment; and

(34) FIG. 30 is a perspective view of three extension arms provided as a kit to the user, one of which is to be used by the user to assemble a windshield mount according to the invention.

DETAILED DESCRIPTION

(35) As used herein, “front” and “forward” connote a direction toward the windshield and the front of a vehicle, and “rear” and “rearward” connote a direction away from the windshield and toward the vehicle interior.

(36) A personal electronic device windshield mount is indicated generally at 100 in FIG. 1. The mount is useful for holding any of a number of relatively small, flat, typically handheld electronic devices, including cell phones, tablets, audio players, map displays, personal digital assistants and

the like. The invention will be described in conjunction with a mobile phone, also called a “smart phone” or a cell phone. These phones typically are rectangular, have heights that greatly exceed their widths, and thicknesses that are small fractions of their widths. Recent cell phones have a display that takes up the entire front surface of the phone.

(37) The mount **100** has a base **102** to which are articulably mounted a first (right) jaw **104** and a second (left) jaw **106**. Jaw **104** has a support plate **108**, while jaw **106** has a support plate **110**. In the illustrated open or release configuration, support plate **110** is residing in a first plane, while support plate **108** is residing in a second plane that is rearwardly and inwardly tilted relative to the first plane. The support or back plates **108** and **110** support a back surface of the cell phone. In the illustrated embodiment, the rear surface of base **102** and plate **110** of left jaw **106** substantially occupy planes that are parallel to each other. In the “open” or “release” configuration, the held device (not shown in this figure) may be removed in a rearward direction orthogonal to the plane of support plate **110**.

(38) In the illustrated embodiment, jaws **104** and **106** move with respect to base **102** in different ways. Jaw **106** may laterally (that is, side-to-side or in a direction at right angles to the front and rear directions) slide toward and away from jaw **104**, but will always reside in the same plane. Jaw **104** does not slide in this embodiment, but instead rotates about an axis that is parallel to the first and second planes. In this embodiment, the jaw **106** slides with respect to base **102**, and may be affixed to base **102**, after slidable adjustment, by means of a clamping bolt or screw **112**. Jaw **104**, on the other hand, is hinged to or pivots relative to the base **102**.

(39) In other embodiments, the positions of rotating jaw **104** and sliding jaw **106** could be reversed, so that the rotating jaw is disposed on the left side, while the sliding jaw is disposed on the right side. A mount with such reversed jaw positions may be more suitable for left-handed users, as the installation of the device into and extraction of the device from the mount may be done with one hand. In another embodiment, two rotating jaws could be provided. In even another alternative embodiment, one jaw could be entirely fixed while the other jaw both rotates and slides.

(40) The base **102** is spherically, or otherwise rotationally, mounted to a rearward end of an elongate attachment arm **114**. The attachment arm **114** extends forwardly from the base **102** to a surface affixation member **116**, which may be a suction cup suitable for attachment by partial vacuum to a glossy surface such as the interior surface of a windshield. In view of the fact that windshields tilt upwardly and rearwardly by varying degrees based on the make and model of the vehicle, different models of the mount **100** could be provided with any of a number of arms **114** of different lengths, and/or different surface affixation members **116** that make different angles to the arms **114**.

(41) FIGS. **18-19** are perspective details of a representative attachment arm assembly **114** which could be used with the invention, as joined to a suction cup assembly **116** which also could be used with the invention. Attachment arm assemblies and suction cup assemblies other than those illustrated could be used, such as those described in FIGS. **21-30**. FIG. **20** is a sectional view of the attachment arm and suction cup assemblies shown in FIGS. **18-19**. The exploded view of FIG. **4** also supplies details of the arm assembly **114** and the suction cup assembly **116**.

(42) The rear end of the illustrated attachment arm assembly **114** takes the form of a cup or partial sphere **400** (FIG. **4**), the rearwardly facing concave surface of which will act as a portion of a ball joint, as will be later described. A knob **418** for a ball joint clamp (further described below) is housed within a space **700** bounded by a first leg **702** and a second leg **704**. Legs **702** and **704** are spaced apart from each other and are long enough to house the knob **418** even where knob **418** has been unscrewed on its bolt shaft (described below) to a loose position. Space **700** intentionally is open to the top and bottom so as to permit the fingers of the user to tighten or loosen knob **418**.

(43) The rear ends of legs **702** and **704** are joined to the cup **400**, as by being integrally molded with it. Legs **702** and **704** are parallel to each other and to an axis of the arm assembly. The legs **702** extend forwardly to and are joined to a transverse plate **706**. A further section **708** of the arm

assembly **114** extends forwardly from the plate **706** to a pivot housing **710**. Section **708** can take any convenient form. In the illustrated embodiment, the section **708** consists of two spaced-apart parallel members **712** and **714**. The members **712** and **714**, and legs **702** and **704**, may be wider in a vertical direction than they are in a horizontal direction, thereby better supporting the combined weight of the cell phone mounting bracket and cell phone mounted on the rearward end of the arm. (44) The pivot housing **710** forms a transverse bore that accepts the shaft of a carriage bolt **716**. A knob **718** is screwed onto the shaft threads of the carriage bolt **716**. The suction cup assembly **116** may include two parallel, spaced-apart wings **800** and **802**. Rearward ends of the wings **800** and **802** have circular holes through them that each accept the shaft of the carriage bolt **716**. Wing **800** is disposed between the head of carriage bolt **716** and pivot housing **710**, while wing **802** is disposed between the pivot housing **710** and the knob **718**. Housing **710** and bolt **716** form a pivot around a transverse axis, allowing the angular adjustment of arm assembly **114** with respect to suction cup assembly **116**.

(45) As best seen in FIG. **19**, the wings **800** and **802** are parallel to each other, are transversely spaced apart and are much taller than they are wide. Wings **800** and **802** extend forwardly until they are joined to a suction cup dish member **804**. The dish member is hollow in an upward-forward direction (FIG. **20**) and is rigid. A plate **806** extends forwardly from wing **800** and downwardly from dish **804**, and a plate **808** extends forwardly from wing **802** and downwardly from dish **804**. The parallel plates **806** and **808** define therebetween a slot **810** that receives a camming lever **812**. The plate **806** has a forwardly open u-shaped surface **814** that acts as a camming surface against which a cam **816** is in contact. Cam **816** is formed to transversely outwardly extend from the general vertical surface of camming lever **812**, so as to interact with camming surface **814**. The right plate **808** has a similar camming surface **809** formed in it (FIG. **3**), and a cam **817** similar to cam **816** extends from the right face of camming lever **812** to interact with it.

(46) As seen in FIG. **20**, a lower end of the camming lever **812** has a hole therethrough that accepts a pin **818**. The pin **818** in turn is threaded through hole(s) of a bottom end of a rod **820**. The rod **820** is positioned substantially along the axis of dish **804**, and extends upwardly and forwardly through a hole in dish **804** until it is joined by any suitable means a rearward surface of an elastomeric disk **822**. A conical spring **830** (FIG. **4**) may be inserted to be coaxial with and disposed around the rod **820**, with a larger-diameter end thereof facing the disk **822** and a smaller-diameter end thereof contacting the inner surface of dish **804**.

(47) In FIGS. **18-20**, camming lever **812** is in a closed position, in which an upper surface **824** of the disk **822** adheres by partial vacuum to the inner side of a glass windshield or other glossy surface. In the closed position, the rod **820** is displaced downwardly and rearwardly along the suction cup axis. It is forced into this position by the action of cam **832** against the lower/rearward surface **834** of the dish member **804**. In an alternative embodiment, the displacement is caused or assisted by cam **816** acting against camming surface **814**. Prior to the attachment by the user to the windshield, the user rotates camming lever **812** downward and rearward, to the position shown in FIG. **4**. This pushes the rod **820** upwardly and forwardly, and upwardly and forwardly displaces the disk **822**, making it flat or more shallow. After the disk **822** has been placed against the windshield, a free end **826** of the camming lever **812** is swung upwardly and forwardly, displacing the rod **820** rearwardly and downwardly and creating a partial vacuum between surface **824** and the windshield. A tab **828** may radially extend from the disk **822** as an aid to later remove the suction cup assembly **116** from the windshield.

(48) Returning to FIG. **1**, a side plate **118** of the left jaw **110** extends rearwardly from a left end **120** of the second support plate **110** to an end **122** of the side plate **118**. A lip **124** of the left jaw **110** extends from end **122** inwardly, in parallel to support plate **110**, to a lip end **126**. The second support plate **110**, side plate **118** and lip **124** form a channel **128** that in combination cages or holds a left lateral side of the cell phone.

(49) Similarly, a side plate **130** of right jaw **104** extends rearwardly from a right end **132** of the first

support plate **108** to a rearward end **134**. A right lip **136** extends from end **134** inwardly, in parallel to support plate **108**, and generally toward channel **128**, to a lip end **138**. The first support plate **108**, side plate **130** and lip **136** form a channel **140** which, in a first or “closed” configuration, will cage a right lateral side of the cell phone.

(50) In the top view of FIG. 2, there can be seen a bottom plate **142** that is joined to second support plate **110**, second jaw side plate **118**, and lip **124**. Bottom plate **142** supports part of the weight of the cell phone being held. Similarly, a bottom plate **144** is joined to first support plate **108**, right jaw side plate **118** and lip **136**. Bottom plate **144** also supports part of the weight of the cell phone, in both the open configuration shown and in the closed configuration. Bottom plates **142**, **144** can be considered to be horizontal portions of the respective left and right channels **128** and **140**.

(51) A first external finger tab **200** is affixed to, and rightwardly extends from, the base **102**. A second external finger tab **202** is affixed to, and rightwardly extends from, side plate **130** of pivoting jaw **104**. The finger tab **200** may reside in a plane parallel to the first plane, which then will be orthogonal to the rearward direction. The finger tab **202** may reside in a plane parallel to the plane occupied by first support plate **108**, and will rotate around axis X, with support plate **108**, as a unit. In the illustrated embodiment, the finger tab **200** is integrally molded as a part of base **102**, while finger tab **202** is integrally molded as a part of right jaw **104**. Arm **114**, base **102**, left jaw **106** and right jaw **104** may be separately injection-molded using a tough plastic such as ABS. The right jaw **104** is an example of a cell phone holding element that can assume a closed configuration in which the cell phone is securely held, and an open configuration in which the cell phone may be easily removed from the mount by the user.

(52) The finger tabs **200** and **202** are sized and positioned to be pinched together with the thumb and one finger of one hand of the user, such as the right hand. The user applies pressure with the thumb to a rearward surface of tab **202**, and applies pressure with e.g. the index finger to a forward surface of tab **200**. The moment arms to these tab surfaces are such that a user will easily be able to overcome the detenting force keeping the first jaw in the closed configuration (described below), and rotate jaw **104** to the open configuration shown, which also is detented.

(53) In FIG. 3, a cell phone C has been mounted to the interior surface of a windshield W by means of the mount **100**. Mount **100** is in the closed configuration in which the cell phone C may not be removed from the mount **100** in a rearward direction, i.e., horizontally away from windshield W. Since the windshield W extends over and rearwardly relative to the cell phone C, the cell phone C may not be removed from mount **100** in an upward direction, either. In the closed configuration, a left lateral side **300** of cell phone C is caged by left jaw channel **128**, while a right lateral side **302** of cell phone C is caged by right jaw channel **140**. In use, jaws **104** and **106** are open in a vertical direction. Except where windshield W or like surface obstructs this movement, the cell phone C is freely insertable into and removable out of the jaws **104** and **106** in the vertical direction.

(54) Contemporary cell phones have a power/communications port in the center of a bottom side **304** of the cell phone. A plug **306** such a firewire plug is inserted into this port. To accommodate plug **306** and the flexible cable **307** attached to it, and as is best shown in FIG. 6, a U-shaped bottom opening **600** persists between left jaw **106** and right jaw **104** even when the jaws **106**, **104** are in their minimum-width position. The opening **600** is defined by an inner margin **602** of left bottom plate **142** and, continuing from margin **602**, an inner margin **604** of the right bottom plate **144**. The provision of opening **600** means that the cell phone C (FIG. 3), as having a power/communications cable **307** plugged into it, may be installed into or removed from mount **100** as a unit.

(55) As shown in FIGS. 4 and 4A, the rearward end of attachment arm **114** takes the form of a partial sphere **400**. The concave interior surface of sphere **400** fits to the front spherical surface **402** of the base **102**. A spherical member **404** rides on a rear spherical surface **406** of the base **102** (FIG. 4A). A square portion **408** of a screw shaft **410** is inserted, in a frontward direction, through a square hole **412** in the spherical member **404**. The square hole **412** prevents the spherical member

404 from spinning on surface **402** upon tightening of knob **418**. Threads **414** of screw shaft **410** will protrude forwardly beyond the spherical member **404**. The screw shaft **410** is then inserted through hole **416** of the partial sphere **400**. A knob **418**, which may have a metal threaded insert, is then threaded onto exposed threads **414**. Parts and surfaces **400-418** form a ball and socket joint. The arm **114** may be rotated on surface **402** to any of a number of positions relative to the base **102**, and the selected position can then be clamped in place with knob **418**.

(56) The left jaw **106** has an elongate, laterally extending slot **420**. A frontwardly recessed region **422** (FIG. 5), with rear-to-front walls that are parallel to the slot **420**, receives a hex head **424** of the clamping screw **112**. A shaft **426** (FIG. 4) of the clamping screw **112** is inserted through slot **420** and then through hole **428** in the base **102** (FIG. 4A). A knob **430**, which may have a threaded metal insert, is then screwed to the exposed threads of shaft **426** to clamp in place the left jaw **106** to the base **102**.

(57) A pair of rails **432**, **434** (FIG. 4) are formed on a front surface **435** of the jaw **106**. Rails **432** and **434** are elongate in a lateral direction, are parallel to each other, and fit within respective channels **436** and **438** of the rear surface of base **102** (FIG. 4A). The rails **432** and **434** and the channels **436** and **438** help constrain jaw **106** to a laterally sliding movement relative to base **102**.

(58) A cylinder **440** (FIG. 4) contains a helical detenting spring **441** that spring-loads a detenting ball **1000** (FIG. 10) to partially protrude out of the cylinder **440**. The cylinder **440** is press-fit into a socket **442** (FIG. 4A) opening on a right side **444** of the base **102**. The socket **442** has an annular recess or countersink **446** that receives a flange **448** of the cylinder **440**.

(59) As seen in FIG. 12, the right jaw **104** has an upper pin **450** and a lower pin **452** that are preferably integrally molded with the jaw **104**. Pin **450** has a shaft-bearing member **453** that extends forwardly from a forward surface **454** of the jaw **104**. A shaft **456** of pin **450** downwardly extends from an end of shaft-bearing member **453** to be received within upper socket **458** (FIG. 4A) formed at the right side **444** of the base **102**. A shaft-bearing member **457** (FIG. 12) of pin **452** extends forwardly from surface **454** at a position downwardly displaced from pin **450**. A shaft **460** of pin **452** downwardly extends from an end of shaft-bearing member **457** to be received within lower socket **462** formed at the right side **444** of the base **102** (FIG. 4A). The bores of sockets **458** and **462** are disposed on axis X and permit pin shafts **456** and **460**, and the rest of jaw **104**, to rotate around axis X.

(60) As seen in FIG. 5, when the mount **100** is in its narrowest-width position, a right margin **500** of support plate **110** mates with or abuts a left margin **502** of support plate **108**. The edges of hex bolt head **424** abut the left end **504** of slot recess **422**. Slot **420** and recess **422** extend rightwardly to respective right ends **506** and **508**. In order to accommodate a longer slot **420**, and therefore an enhanced range of cell phone widths capable of being mounted, the margins **500**, **502** are not straight. In another embodiment, they could be. In the illustrated embodiment, margin **502** creates a first, upper interior tab **510** in the upper region of support plate **108**, and a second, lower interior tab **512** in a lower region of the support plate **108**. As will be further described below, it is these tabs **510**, **512** which will contact the back of the cell phone to push it out of a caged condition, and it will be these tabs **510**, **512** which will be pressed frontward by the back of the cell phone to induce the rotation of jaw **104** to the closed configuration. The non straight right margin **500** of support plate **110** creates a rightward-extending central lobe **514** that surrounds the right ends **506**, **508** of respective slot **420** and recess **422**.

(61) In FIG. 7, the left jaw **106** has been slid to its maximum-width position, in which the left channel **128** is farthest away from right channel **140**. The edges of hex bolt head **424** abut the right end **608** of slot recess **422**. In other embodiments, the maximum distance between channel **128** and channel **140** could be further increased, so as to allow the installation of the cell phone in a "landscape" configuration.

(62) In FIG. 9, the jaw **104** has been rotated by an angle θ from a closed configuration, in which side plate **130** is perpendicular to the first plane (occupied by second support plate **110**; see FIG. 8),

to an open configuration, in which lip end **138** has been swung laterally outwardly around axis X. If the radius from axis X to lip end **138** is r , and if the depth of channel **140** is defined by the distance of lip end **138** to the inner surface of side plate **130**, then the sine of angle θ should be at least the ratio of the channel depth to radius r . With the proportions shown, θ should be at least about five degrees. In the illustrated embodiment, θ is about sixteen degrees.

(63) In one embodiment, a detenting structure indexes the open and closed configurations of the jaw **104**. One such detenting mechanism is shown in FIGS. **10-12**. A camming member **1200** (FIG. **12**) forwardly extends from a general front surface **454** of jaw **104**. The camming member has an inwardly-facing convex camming surface **1202** with a first concave detenting surface **1204**, and, immediately forward of or overlapping the first detenting surface **1204**, a second concave detenting surface **1206**. In FIG. **10**, a detenting ball **1000** has been cammed into concave detenting surface **1204** by the spring **441** housed in cylinder **440** (FIG. **4**). This detents jaw **104** to the closed configuration. In FIG. **11**, the detenting ball **1000** has been cammed into concave detenting surface **1206** by the detenting spring **441**. This detents jaw **104** to the open or release configuration. The detenting mechanism, through an audible and tactile sensation, tells the user when he or she has achieved either configuration. Further, the detenting mechanism ensures that the chosen configuration will remain until consciously changed by the user. In an alternative embodiment (not shown), the detenting structure shown in FIGS. **10-12** could be reversed. The spring-loaded ball **1000** could be carried on the pivoting jaw **104**, while a concave-shaped camming surface could be molded as a part of base **102**. In other embodiments, the detenting structure could incorporate another pressuring element such as a leaf spring instead of a spring-loaded ball. The detenting structure creates a resistance to rotation of jaw **104** when the jaw **104** is at either detented position. The user must exert at least a predetermined amount of force to escape either detented position.

(64) FIG. **12** also illustrates upper and lower retention tabs **1208**, **1210** that forwardly extend from the general forward surface **454** of jaw **104**. Upper retention tab **1208** has a terminal barb **1212** while lower retention tab **1210** has a terminal barb **1214**. During assembly of the mount, the barb **1212** of upper retention tab **1208** is snapped forwardly through an upper retention tab hole **464** (FIG. **4A**), while the barb **1214** of lower retention tab **1210** is snapped forwardly through a lower retention tab hole **466**. Once the barbs **1212**, **1214** have been inserted through holes **464** and **466**, they will prevent easy withdrawal of tabs **1208** and **1210**. The purpose of retention tabs **1208** and **1210** is to limit the degree to which jaw **104** can be swung out. Tabs **1208** and **1210** effectively prevent jaw **104** from being swung out beyond the open configuration detented position.

(65) In FIG. **13**, a right lateral side **302** of cell phone C has been “caged” by channel **140**, since jaw **104** is in its closed configuration. In this posture, the lip **136** will prevent the removal of cell phone C in a rearward direction (toward the bottom of the page in this FIGURE).

(66) In FIG. **14**, jaw **104** is in the open or release configuration, by which lip **136** and lip end **138** have been swung out of the way. Lip **136** therefore no longer presents a barrier to the rearward extraction of cell phone C. Further, inner tab **510** (and lower tab **512**, not shown in this FIGURE) are in contact with the back surface **1400** of cell phone C.

(67) This FIGURE illustrates a cell phone C about to be extracted from mount **100**, or a cell phone C about to be installed into a mount **100**. In the former instance, the user has pinched tab **202** toward tab **200**, causing the entirety of jaw **104** to rotate laterally outwardly. Tab **510** (and tab **512**) push on the back surface **1400** to rearwardly displace the cell phone C, making cell phone C more extractable by one hand. In the latter instance, the user is about to push tab **510** (and tab **512**) in a frontward direction through the back surface **1400** of cell phone C. A user is thereby able to push tabs **510**, **512**, thereby rotating jaw **104** out of the second detented position back to the first detented position, and swinging in lip **136** to cage the lateral right side **302** of the phone C.

(68) FIGS. **15A-15C** and FIG. **16** illustrate steps in a personal electronic device mounting process made possible by the invention. At step **1598**, the user loosens the jaw-clamping knob **430** and opens the jaws by sliding jaw **110** leftward to its widest position. At step **1600**, the user uses inner

tabs **510**, **512** to rotate jaw **104** to the closed configuration, either by pressing them with his or her fingers or using the back surface **1400** of the cell phone. This last method will automatically cause the phone to be inserted at **1602**. Otherwise, the user vertically drops the phone into the jaws **104**, **106** at this step. See FIG. **15A**.

(69) At step **1604**, the user slides jaw **106** toward or away from jaw **104**, thereby adjusting or “sizing” the width between jaws **104** and **106** to the width of the cell phone. At step **1606**, the user pinches the external tabs **200**, **202** to rotate jaw **104** to the open configuration, and removes the phone (**1608**). See FIG. **15B**. Alternatively, the user simply vertically lifts the phone out of the jaws.

(70) Now that the mount **100** is again empty, at step **1610** the user affixes the mount **100** to a surface, such as the interior surface of a windshield. This can be done by actuating the suction cup assembly **114**. If an articulated arm assembly such as the one shown in FIGS. **18-20** is used, the user may also adjust the angle of the arm assembly **114** relative to the axis of the suction cup assembly **116**. If a telescoping arm assembly such as the one shown in FIGS. **22-29** is used, the user may also adjust the arm length at this time.

(71) Preliminary to step **1612**, the mount **100** must be in the open configuration and empty. In this condition, the user displaces the cell phone in a forward direction, until tabs **510** and **512** are contacted. When contact with the cell phone back surface **1400** is made (FIG. **14**), further pushing by the user will cause (**1614**) the right jaw **104** to rotate closed (FIG. **13**). The pushing force exerted by the user must be at least a predetermined force for overcoming the resistance created by the detent structure. Responsive to this, the mount **100** is now in a closed configuration, and securely holds the phone. If an adjustment of the cell phone viewing plane is necessary, this is done at step **1616**, following which the viewing plane is set by tightening knob **418**.

(72) The mounted phone is now ready for use. When the user wishes to remove the phone from the mount, at step **1700** (FIGS. **15B** and **17**) the user pinches external finger tab **202** toward finger tab **200**, which may be accomplished with one hand. Responsive to this force, the jaw **104** leaves the detented closed configuration and rotates to the open configuration at step **1702**. This uncages the right side of the phone at step **1704**. The user may then remove the cell phone with one hand at step **1706** (FIG. **15C**). FIG. **15C** shows this step being accomplished by the left hand, but it could as or more easily be accomplished with the right hand.

(73) Now that the mount has been set up and affixed to the windshield, the user need only perform step **1612** to re-install the phone into the mount, as the width between the channel jaws has already been customized to the width of the phone.

(74) A further embodiment of a windshield mount for a personal electronic device is indicated generally at **2200** in FIG. **22**. Mount **2200** includes a surface affixation member or windshield mounting component **116**, which in the illustrated embodiments is a suction cup assembly and which has been described in detail above. The mounting component **116** is rotatably attached to a forwardmost arm component **2202** of a telescoping arm assembly **2204** by means of carriage bolt **716** through pivot housing **710** (FIG. **23**). An intermediate arm component **2206** slides in and out of a channel (described below) formed in forward arm component **2202** and is affixed thereto by means of a flat-headed bolt (described below) and a knob **2208**. A rearwardmost arm component **2210** slides in and out of a channel (described below) formed in component **2206** and is affixed thereto by means of a further bolt (described below) and a knob **2211**. Component **2210** terminates in a partial sphere **400** with a rearwardly facing concave surface. Parts **400-418** (see FIG. **4**) form a ball and joint assembly **2212** that has been described above. The ball and joint assembly **2212** joins the arm assembly **2204** to a personal electronic device mounting bracket **2214**, one embodiment of which has been described in detail in conjunction with FIGS. **1-16**.

(75) The illustrated embodiment of the telescoping arm assembly **2204** has three arm components. Other embodiments may include only two telescoping components, or may have more than three.

(76) An elastomeric bumper **2216** is mounted on a forward portion **2502** of the arm component

2210. Bumper **2216** may be formed of a thermoplastic elastomer (TPE) and may be disposed at any of various user-selected positions along arm component **2210**.

(77) A total length of the arm assembly **2204** from bolt/axle **716** to ball and socket assembly **2212**, and from a front end of assembly **2204** to a rear end thereof, may be adjusted by the user by sliding one or both of the arm components **2206**, **2210** into and out of the channels of the respective arm components to their immediate front. FIG. **23** depicts arm assembly **2204** in its full extension, in which the length of the arm assembly **2204** between its front end and its rear end is maximized.

This may be done where the mount **2200** is installed in a large vehicle with a relatively large depth of a dashboard **2300** between windshield **W** and a rearward dashboard edge **2302**. For vehicles with less-deep dashboards, less of the lengths of the arm components **2206** and **2210** need to be used.

(78) The bumper **2216** rests on a top surface **2304** of the dashboard **2300**, near its rearward edge **2302**. A rearward portion **2306** of the arm component **2210** extends rearwardly beyond the dashboard edge **2302**, and the bracket **2214** is suspended in space to the rear of edge **2302**. As shown in FIG. **21**, this brings the cell phone bracket **2214** within a maximum reach envelope **2100** of a driver **D**, making the cell phone or other personal electronic device carried by bracket **2214** more safely and easily operable by driver **D**, particularly where driver **D** is restrained by a seat belt. The cell phone holding device, as including the telescoping arm assembly **2204**, permits the positioning of the cell phone **C** very close to the top surface **2304** of dashboard **2300**, and creates minimal view restriction. The resting of the bumper **2216** on the dashboard surface **2304** increases arm stability and capacity to bear the weight of cell phone **C**, which can be heavy.

(79) FIGS. **24** and **25** best illustrate how the different extension arm components **2202**, **2206**, **2210** slide in and out of, and are connected to, each other. Arm component **2202** has a lipped channel **2400** which slidably receives the body of arm component **2206**. Arm component **2206** has a slot **2402** which extends for most of its length. A flat-headed bolt **2404** is inserted through slot **2402** and through a hole **2403** (FIG. **26**) in arm component **2202**. The shaft **2405** of bolt **2404** is threadably engaged with knob **2208** (FIG. **23**). A head **2406** of bolt **2404** has flat sides **2408** that ride within a groove **2410** that is a little wider than slot **2402**. The depth of bolt head **2406** is less than the depth of groove **2410**, so as not to interfere with the movement of arm component **2210** (FIG. **25**).

(80) As seen in FIG. **24**, arm component **2206** may be slid out of lipped channel **2400** until a front end **2412** of the slot **2402** abuts the shaft **2405** of bolt **2404**. Arm component **2206** may be slid into the lipped channel **2400** until a rear end **2414** of slot **2402** abuts the shaft **2405** of bolt **2404**. The user is thus free to select a position of arm component **2206** relative to arm component **2202** anywhere between and including slot ends **2412** and **2414**. Once a position is selected, the user tightens knob **2208** (FIG. **22**) onto bolt **2404** to clamp the component **2206** to component **2202**.

(81) The articulation of arm component **2210** with arm component **2206** is similar to that between arm components **2202** and **2206**, and is illustrated in FIG. **25**. Arm component **2206** has a lipped channel **2500** which slidably receives a front portion **2502** of the arm component **2210**. The rearwardmost arm component **2210** has a slot **2504** that extends between a front end **2506** and a rear end **2508** thereof. A bolt **2510** is inserted through slot **2504**. A threaded shaft **2512** of the bolt **2510** is threaded through a hole **2514** in the component **2206** and threadably engages with the knob **2211** (FIGS. **23** and **26**). A head **2516** of the bolt **2510** (FIG. **25**) has flat sides **2518** that slidably engage with a groove **2520**, the width of which is a little wider than that of slot **2504**.

(82) Front portion **2502** of component **2210** may be slid into the lipped channel **2500** until rear slot end **2508** abuts bolt shaft **2512**. Contrariwise, front portion **2502** of component **2210** may be slid out of the lipped channel **2500** until front end **2506** abuts the bolt shaft **2512**. The user may select either of these extreme positions, or any position in between, and may then tighten knob **2211** onto bolt shaft **2512** to clamp component **2210** to component **2206** at the desired location. Extending more of front component **2210** out of intermediate component **2206** expands the range of positions at which bumper **2216** may be sited.

(83) Rearwardmost arm component **2210** further has a rear portion **2522** which rearwardly extends

from front portion **2502**. A transverse plate **2524** is joined to a rear end of the portion **2502**. From plate **2524**, two spaced-apart opposed legs **2526** and **2528** extend rearwardly to partial sphere **400**. The space rearward of plate **2524**, inward of legs **2526** and **2528**, and frontward of partial sphere **400**, accommodates the ball joint clamping screw knob **418** (FIGS. **22** and **23**). As seen in the exploded view of FIG. **26**, each of the knobs **2208**, **2211** may include threaded metal inserts **2600**. (84) With reference to FIG. **27**, the bumper **2216** may be fitted around the front portion **2502** of component **2210** anywhere along the exposed part of slot **2504**. A downwardly extending finger **2800** of bumper **2216** (FIG. **28**) fits within slot **2504**. Once installed onto portion **2502**, the position of bumper **2216** along slot **2504** may be adjusted the user by sliding bumper **2216** on the exposed part of portion **2502**.

(85) Bumper **2216** may have a substantially constant cross-section along its axial depth (that is, along its dimension that is aligned in use with the arm assembly **2204**). In one embodiment, the bumper has a hollow inner shape **2802** (FIG. **28**) that substantially, but not completely, conforms to the cross-sectional shape **2804** of arm component front portion **2502**. The finger **2800** forms part of this inner shape **2802**. The inner shape has left and right laterally extending grooves **2806**, **2808** that are intentional departures from cross-sectional shape **2502**.

(86) The bumper **2216** has a cross-sectional outer locus or periphery **2810**, shown in broken line, that is outwardly spaced from its inner shape **2802**. In use, one of an upper portion **2812** of the periphery **2810**, and a lower portion **2814** of the periphery **2810**, will be in contact with dashboard surface **2304**. The bumper **2216** therefore suspends arm component **2502** off of dashboard surface **2304** and supports a majority of the weight of the cell phone mounting bracket **2214**, and cell phone C, both of which are disposed rearwardly of bumper **2216**.

(87) In the illustrated embodiment, an outer shape **2811** of the bumper is not simple but rather is heavily reticulated. To form the outer shape **2811**, the upper and lower peripheries **2812**, **2814** are interrupted by a series of deep grooves **2816**, each of which extends from periphery **2810** to a respective bottom near the inner shape **2802**. The grooves **2816** enhance the moldability of bumper **2216** and also increase its flexibility and ability to absorb shock. The grooves **2816** define a plurality of fingers **2817** therebetween.

(88) The bottoms of certain of these grooves **2816** cooperate with the inner shape **2802** to form a series of living hinges. In the illustrated embodiment, there at least six of these. A bottom **2818** of upper left groove **2820** is disposed near the bottom of lateral groove **2806**, creating a living hinge **2822** therebetween. A bottom **2824** of lower left groove **2826** is also disposed near the bottom of inner lateral groove **2806**, creating a living hinge **2828** therebetween. A bottom **2830** of upper right groove **2832** is disposed near the bottom of right lateral groove **2808**, creating a living hinge **2834** therebetween. A bottom **2836** of lower right groove **2838** is disposed near the bottom of right lateral groove **2808**, creating a living hinge **2840** therebetween. A bottom **2842** of upper central groove **2844** is close to an upper left corner **2846** and to an upper right corner **2848** of the inner shape **2802**, respectively creating living hinges **2850** and **2852** therebetween. Upper left corner **2846** is formed at the junction of a left upper surface **2845** of inner shape **2802** and slot finger **2800**, while upper right corner **2848** is formed at the junction of a right upper surface **2847** of inner shape **2802** and finger **2800**. The finger **2800** divides the left upper surface **2845** from the right upper surface **2847**.

(89) The bumper **2216** substantially surrounds, in a cross-sectional plane, the arm component portion **2502**. But bumper **2216** has an opening **2858** that extends from the periphery **2810** to the inner shape **2802**. Opening **2858** is defined by a left wall **2854** which, in the illustrated embodiment, is spaced from a right wall **2856**; in other embodiments walls **2854** and **2856** may have no space between them in an unflexed condition. Opening **2858** separates a left side or portion **2855** of the bumper **2216** from a right side or portion **2857**. Portions **2855** and **2857** are connected together only by a living hinge, or, as in the illustrated embodiment, by multiple living hinges **2824**, **2822**, **2850**, **2852**, **2834** and **2840** arranged in series.

(90) Because bumper **2216** is elastomeric, it may be installed on the arm portion **2502** by prying apart wall **2854** from wall **2856** and inserting therethrough the arm segment portion **2502**. The maximum degree of distortion of bumper **2216** comes as the arm portion **2502** forces apart inner shape shoulder **2860** from inner shape shoulder **2862**, as seen in FIG. **29**. At this point all of the living hinges **2828**, **2822**, **2850**, **2852**, **2834** and **2840** are flexed, permitting this distortion to occur without tearing the bumper. This flexing is also aided by the deep grooves, particularly grooves **2826**, **2820**, **2844**, **2832** and **2838**, as bumper material may be displaced into those grooves during flexure.

(91) After the arm segment portion **2502** passes by shoulders **2860** and **2862**, the elastomeric living hinges **2828**, **2822**, **2850**, **2852**, **2834** and **2840** will urge the bumper **2216** to assume its unflexed shape, as seen in FIG. **28**. The arm segment front portion **2502** will then be seated within inner shape **2802** and most of the walls of inner shape **2802** will be adjacent to respective walls of the arm segment portion **2502**.

(92) In another embodiment (not shown), bumper **2216** may be affixed to the arm component **2210** by means other than, or in addition to, elastically snapping the bumper body around the arm by the use of living hinge(s). For example, bumper **2216** could be affixed to the arm component **2210** by means of a screw, pin or a non-living hinge or closure that would simply be closed once the bumper was positioned at the desired location.

(93) In one embodiment, a kit of different attachment arms and arm assemblies is supplied to the consumer. Once such kit is shown in FIG. **30**. The kit **3000** includes a short arm **114**, identical to the arm **114** depicted in FIGS. **1-4** and **15A-15C**. The short arm **114** has a predetermined length between a front end **3001** and a rear end **3003** thereof. The kit **3000** further includes a longer, but still one-piece, arm **3002**, which differs from arm **114** only in that the length between a rear end **3004** and a front end **3006** thereof is longer. In particular, parallel members **3007** and **3009** are longer than corresponding arm members **712** and **714**, all other elements of arms **114**, **3002** being similar to each other. The final element of kit **3000** is arm assembly **2204** and its three arm components **2202**, **2206** and **2210**, all previously described. A total length of assembly **2204** between a front end **3008** and a rear end **3010** thereof is variable, but in any event is longer than the length between ends **3004**, **3006** of second arm **3002**. The user selects one of these arms or arm assemblies for use in assembling it to a windshield attachment unit **116** and a device mount **2214**, thereby creating a complete mount.

(94) In summary, a telescoping extension arm assembly has been provided that brings the mounted personal electronic device within the maximum reach envelope of the driver. A bumper is used to stabilize the mount and create a point of contact near the dashboard edge, supporting the device mounting bracket rearwardly from the bumper and the dashboard edge in cantilever fashion. A kit of arms may be provided to the consumer for the consumer to selectively use in assembling a complete windshield phone mount.

(95) While illustrated embodiments of the present invention have been described and illustrated in the appended drawings, the present invention is not limited thereto but only by the scope and spirit of the appended claims.

Claims

1. A mount for a personal electronic device, the mount comprising: a mounting component for affixation to a rear surface of vehicle windshield; a device mounting bracket for holding the personal electronic device; an elongate arm assembly joined to and extending between the mounting component and the device mounting bracket, a total length of the elongate arm assembly extending, in use, beyond a rearward margin of a vehicle dashboard, such that the device mounting bracket is positioned rearwardly of the rearward margin of the vehicle dashboard, a first arm component of the arm assembly positioned, in use, over the rearward margin of the vehicle

dashboard, the first arm component having a cross-sectional shape; and a bumper integrally molded of elastomeric material and substantially surrounding the cross-sectional shape of the first arm component, a hollow inner shape of the bumper substantially conforming to the cross-sectional shape of the first arm component, a bumper outer periphery spaced from and surrounding the inner shape, an opening extending from the outer periphery to the inner shape, the opening having first and second opposed sidewalls disposed at a first width from each other when the bumper is in an unflexed condition, at least one outer groove inwardly extending toward the inner shape from the outer periphery and having a groove bottom near the inner shape, at least one living hinge formed between the groove bottom and the inner shape, such that prying force flexes the at least one living hinge and moves the first sidewall away from the second sidewall such that the first sidewall is disposed at a second width from the second sidewall which is greater than the first width, the opening then permitting the insertion of the cross-sectional shape of the first arm component into the inner shape of the bumper, the first sidewall returning to be disposed at the first width from the second sidewall once the prying force is removed; the bumper positioned on the first arm component near the rearward margin of the vehicle dashboard, whereby the bumper and not the first arm component will contact the dashboard near the rearward margin thereof.

2. The mount of claim 1, wherein a plurality of outer grooves inwardly extend from the outer periphery toward the inner shape, each of the outer grooves having a bottom disposed near the inner shape, a plurality of living hinges formed between ones of the bottoms of the outer grooves and the inner shape.

3. The mount of claim 1, wherein the arm assembly includes a second arm component, the first arm component slidable with respect to the second arm component such that a length of the arm assembly between the mounting component and the device mounting bracket may be adjusted.

4. The mount of claim 3, wherein the arm assembly further includes a third arm component, the second arm component slidably attached to the third arm component, the first arm component slidably attached to the second arm component, a forward end of the third arm component joined to the mounting component, a rear end of the first arm component joined to the device mounting bracket.

5. The bumper and arm assembly of claim 1, wherein the bumper is slidable to any of a plurality of user-selected positions along the major portion of the length of the arm.

6. A bumper for supporting an elongate structural member above a support surface, the bumper comprising: a body integrally molded of an elastomeric material; a hollow inner shape formed in the body, the shape substantially conforming to a predetermined cross-sectional shape of the structural member; the body having an outer periphery substantially surrounding the inner shape in cross section and spaced from the inner shape, a portion of the outer periphery contacting the support surface when the bumper is mounted around the structural member and when the bumper is being used to support the structural member above the support surface; a plurality of outer grooves opening on the outer periphery and extending inwardly toward the inner shape, each outer groove having a bottom, at least one living hinge formed between one of the bottoms of the outer grooves and the inner shape; an opening in the body extending from the outer periphery to the inner shape, the opening spacing a first side of the body from a second side of the body by a first width when the body is in an unflexed condition; the bumper adapted to being installed around the cross-sectional shape of the structural member by exerting prying force between the first side and the second side so that the first side is forced away from the second side, such that the first side is spaced from the second side by a second width larger than the first width, the at least one living hinge flexing responsive to the prying force, the cross-sectional shape of the structural member then being inserted through the opening and seated within the inner shape of the body of the bumper, the first side returning to a position at the first width from the second side when the prying force is removed.

7. The bumper of claim 6, wherein the inner shape has a bottom opposed to the opening, a left side

wall upwardly extending from the bottom and a right side wall upwardly extending from the bottom and spaced from the left side wall, a left inner groove leftwardly extending toward the periphery from the left side wall and having a left inner groove bottom, a right inner groove rightwardly extending toward the periphery from the right side wall and having a right inner groove bottom, the outer grooves including a first outer groove and a second outer groove, a bottom of the first outer groove being near the left inner groove bottom, a first living hinge formed by the proximity of the bottom of the first outer groove to the left inner groove bottom, a bottom of the second outer groove being near the right inner groove bottom, a second living hinge formed by the proximity of the bottom of the second outer groove to the right inner groove bottom.

8. The bumper of claim 7, wherein the outer grooves further include a third outer groove with a bottom disposed near the left inner groove bottom and a fourth outer groove with a bottom disposed near the right inner groove bottom, a third living hinge formed by the proximity of the bottom of the third outer groove to the left inner groove bottom, a fourth living hinge formed by the proximity of the bottom of the fourth outer groove to the right inner groove bottom.

9. The bumper of claim 6, wherein the outer periphery has an upper portion, the outer grooves including a central outer groove extending downwardly from the upper portion, the inner shape having a left upper surface, a right upper surface and a downwardly extending finger dividing the left upper surface from the right upper surface, the left upper surface and the finger forming a left upper corner and the right upper surface and the finger forming a right upper corner, a bottom of the central outer groove being disposed near the left and right upper corners of the inner shape, a first living hinge formed between the left upper corner and the bottom of the central outer groove, a second living hinge formed between the right upper corner and the bottom of the central outer groove.
